

Gradient Descent

Hoa Vu
CS 663

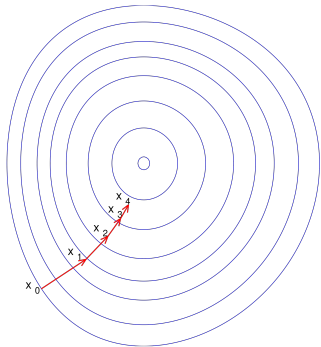
SDSU

April 6, 2026

Maximization and Minimization

Minimize function say $f(x, y) = 2e^{-x^2+1} + y^2$.

Gradient Descent/Ascent



At each step, take a small step in the direction that is steepest descent/ascent.

Gradient Descent

It turns out that the direction of steepest ascent is given by the gradient

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}.$$

For example, $f(x, y) = 2e^{-x^2+1} + y^2$ has the gradient:

$$\nabla f(x, y) = \begin{pmatrix} -4x e^{-x^2+1} \\ 2y \end{pmatrix}.$$

Gradient Descent

To minimize f , we can use the following update rule:

$$\begin{pmatrix} x \\ y \end{pmatrix} \leftarrow \begin{pmatrix} x \\ y \end{pmatrix} - \alpha \nabla f(x, y),$$

where $\alpha > 0$ is the learning rate.

Gradient Descent

$f(x, y) = 2e^{-x^2+1} + y^2$ has the gradient:

$$\nabla f(x, y) = \begin{pmatrix} -4x e^{-x^2+1} \\ 2y \end{pmatrix}.$$

Suppose we start at $(x, y) = (1, 1)$

$$f(1, 1) = 2e^{-1^2+1} + 1^2 = 3.$$

Gradient Descent

Use a learning rate of $\alpha = 0.1$. Then:

$$\nabla f(1, 1) = \begin{pmatrix} -4(1)e^0 \\ 2(1) \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}.$$

The first update is:

$$\begin{pmatrix} x \\ y \end{pmatrix} \leftarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 0.1 \begin{pmatrix} -4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1.4 \\ 0.8 \end{pmatrix}.$$

Then,

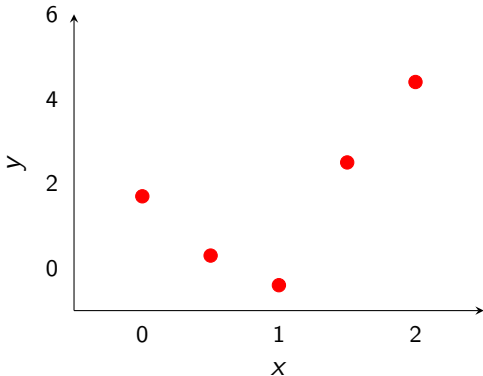
$$f(1.4, 0.8) = 2e^{-0.96} + 0.64 \approx 1.406 < 3 = f(1, 1).$$

Fitting a curve

We have 6 data points (x_i, y_i) :

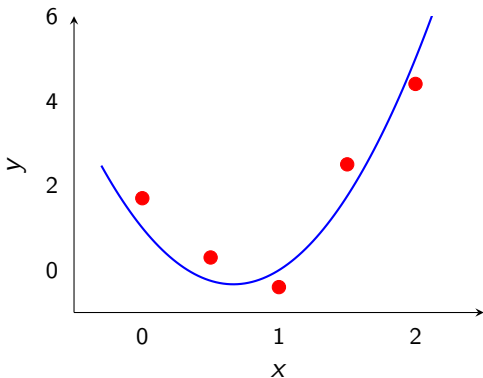
$(0, 1.7), (0.5, 0.3), (1, -0.4), (1.5, 2.5), (2, 4.4)$

We suspect that the data follows a quadratic relationship.



Fitting a curve

Find a quadratic function $y = ax^2 + bx + c$ that best fits the data.



Squared error loss

We can measure how well the curve fits the data using the squared error loss:

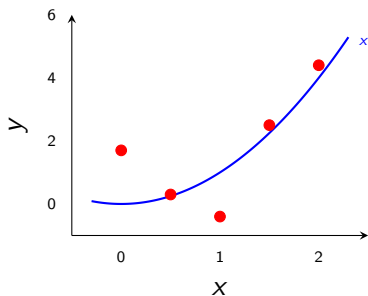
$$L(a, b, c) = \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c))^2.$$

We want to find a, b, c that minimize $L(a, b, c)$.

$$(y_i - (ax_i^2 + bx_i + c))^2$$

is the squared distance from the point y_i to the curve at x_i .

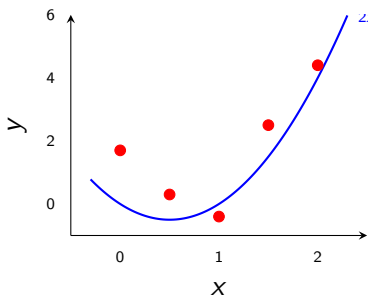
Example: $a = 1$, $b = 0$, $c = 0$



$$f(x) = x^2$$

$(y_i - f(x_i))^2$	
$(1.7 - 0)^2$	$= 2.89$
$(0.3 - 0.25)^2$	$= 0.00$
$(-0.4 - 1)^2$	$= 1.96$
$(2.5 - 2.25)^2$	$= 0.06$
$(4.4 - 4)^2$	$= 0.16$
L	≈ 5.08

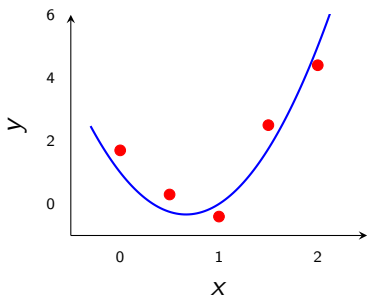
Example: $a = 2$, $b = -2$, $c = 0$



$$f(x) = 2x^2 - 2x$$

$(y_i - f(x_i))^2$	
$(1.7 - 0)^2$	$= 2.89$
$(0.3 - (-0.5))^2$	$= 0.64$
$(-0.4 - 0)^2$	$= 0.16$
$(2.5 - 1.5)^2$	$= 1.00$
$(4.4 - 4)^2$	$= 0.16$
L	≈ 4.85

Example: $a = 3$, $b = -4$, $c = 1$



$$f(x) = 3x^2 - 4x + 1$$

$(y_i - f(x_i))^2$	
$(1.7 - 1)^2$	$= 0.49$
$(0.3 - (-0.25))^2$	$= 0.30$
$(-0.4 - 0)^2$	$= 0.16$
$(2.5 - 1.75)^2$	$= 0.56$
$(4.4 - 5)^2$	$= 0.36$
L	≈ 1.88

Gradient Descent

$$L(a, b, c) = \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c))^2.$$

The gradient of L is:

$$\nabla L = \begin{pmatrix} \frac{\partial L}{\partial a} \\ \frac{\partial L}{\partial b} \\ \frac{\partial L}{\partial c} \end{pmatrix} = -2 \sum_{i=1}^n (y_i - ax_i^2 - bx_i - c) \begin{pmatrix} x_i^2 \\ x_i \\ 1 \end{pmatrix}.$$

Gradient Descent

Therefore, the update rule is:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \leftarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} + 2\alpha \sum_{i=1}^n (y_i - ax_i^2 - bx_i - c) \begin{pmatrix} x_i^2 \\ x_i \\ 1 \end{pmatrix},$$

where $\alpha > 0$ is the learning rate.